Which data lakehouse feature results in improved data quality over a traditional data lake?

A. A data lakehouse stores data in open formats.

B. A data lakehouse allows the use of SQL queries to examine data.

C. A data lakehouse provides storage solutions for structured and unstructured data.

D. A data lakehouse supports ACID-compliant transactions.

Which of the following describes the relationship between Gold tables and Silver tables?

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C Gold tables are more likely to contain a less refined view of data than Silver tables.

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You are working on a data analysis project where you need to transform a DataFrame named 'df_products' from a long format to a wide format for better readability and analysis. The DataFrame contains the following columns: 'product_id', 'product_name', 'category', and 'price'. Your task is to use the Spark SQL 'PIVOT' clause to achieve this transformation, creating separate columns for each category ('Electronics', 'Clothing', 'Food') and calculating the total price for each category. Consider the following constraints: the solution must be scalable to accommodate additional categories without manual updates, and it should efficiently handle large datasets. Which of the following queries meets these requirements? (Choose two correct options if option E is available, otherwise choose one.)

A. SELECT product_id, product_name, category, SUM(price) as total_price FROM df_products GROUP BY product_id, product_name, category

B. SELECT product_id, product_name, SUM(price) as total_price FROM df_products GROUP BY product_id, product_name

C. SELECT product_id, product_name, Electronics, Clothing, Food FROM (SELECT product_id, product_name, category, price FROM df_products) PIVOT (SUM(price) FOR category IN ('Electronics', 'Clothing', 'Food'))

D. SELECT product_id, product_name, category, SUM(price) as total_price FROM df_products PIVOT (SUM(price) FOR category)

E. SELECT product_id, product_name, category, SUM(price) as total_price FROM df_products GROUP BY product_id, product_name, category; SELECT product_id, product_name, Electronics, Clothing, Food FROM (SELECT product_id, product_name, category, price FROM df_products) PIVOT (SUM(price) FOR category IN ('Electronics', 'Clothing', 'Food'))

Which of the following benefits of using the Databricks Lakehouse Platform is provided by Delta Lake?

A The ability to manipulate the same data using a variety of languages

B The ability to collaborate in real time on a single notebook

C The ability to set up alerts for query failures

D The ability to support batch and streaming workloads

E The ability to distribute complex data operations

Explain the role of notebook libraries in a Delta Live Tables pipeline. How do these libraries contribute to the transformation and processing of data within the pipeline, and what are the implications of not including them?

A. Notebook libraries are used for defining the source data locations and the target schema.

B. Notebook libraries contain the transformation logic and data processing steps that are applied to the data as it flows through the pipeline.

C. Notebook libraries are used solely for data visualization and reporting purposes.

D. Notebook libraries are optional components that can be used for additional data validation checks.

You are tasked with identifying how often a temperature sensor exceeds a threshold temperature of 100.00 for each device. Each row in the dataset contains 5 readings collected every 5 minutes. Complete the SQL query by filling in the blanks with the appropriate functions.

Table Schema

- deviceId INT
- deviceTemp ARRAY<DOUBLE>
- dateTimeCollected TIMESTAMP

```sql
SELECT
  deviceId,
  __(__(__ (deviceTemp, i -> i > 100.00)))
FROM devices
GROUP BY deviceId
```

A	SUM, COUNT, SIZE
B	SUM, SIZE, SLICE
C	SUM, SIZE, ARRAY_CONTAINS
D	SUM, SIZE, ARRAY_FILTER
E	SUM, SIZE, FILTER

A data analysis team has noticed that their Databricks SQL queries are running too slowly when connected to their always-on SQL endpoint. They claim that this issue is present when many members of the team are running small queries simultaneously. They ask the data engineering team for help. The data engineering team notices that each of the team's queries uses the same SQL endpoint.
Which approach can the data engineering team use to improve the latency of the team's queries?

A. They can increase the cluster size of the SQL endpoint.

B. They can increase the maximum bound of the SQL endpoint's scaling range.

C. They can turn on the Auto Stop feature for the SQL endpoint.

D. They can turn on the Serverless feature for the SQL endpoint.

Which type of view is not accessible outside the notebook where it is declared?

A Global Temporary View

B View and Temporary View

C View

D Temporary View

E Global Temporary View and Temporary View

A new data engineering team has been assigned to work on a project involving the analysis of existing customer data. To effectively carry out their tasks, the team requires access to the database named 'customers' to inspect the existing tables and their structures. The team operates under its own designated group. Which SQL command should you use to grant the necessary permissions on the entire 'customers' database to the new team?

A GRANT VIEW ON CATALOG customers TO team;

B GRANT CREATE ON DATABASE customers TO team;

C GRANT USAGE ON CATALOG team TO customers;

D GRANT USAGE ON DATABASE customers TO team;

When setting up a Delta Live Table pipeline in Databricks, which field is not mandatory?

A Pipeline Name

B Pipeline Mode

C Target Schema

D Notebook Libraries

E Min & Max Workers when Autoscaling is enabled

When using dashboard tasks in Lakeflow Jobs, what is required for the dashboard to function properly?

A The dashboard must be published and connected to a SQL warehouse

B The dashboard must use only real-time data sources

C The dashboard must be created using Python notebooks only

D The dashboard must be private to the job creator

Which of the following code blocks will remove the rows where the value in column age is greater than 25 from the existing Delta table my_table and save the updated table?

A SELECT * FROM my_table WHERE age > 25;

B UPDATE my_table WHERE age > 25;

C DELETE FROM my_table WHERE age > 25;

D UPDATE my_table WHERE age <= 25;

E DELETE FROM my_table WHERE age <= 25;

When a dataset defined using Delta Live Tables includes an expectations clause with 'ON VIOLATION FAIL', what happens if a batch of data violates these constraints?

A. Records violating the expectation are added to the target dataset and marked as invalid in the event log.

B. Records violating the expectation are dropped from the target dataset and noted as invalid in the event log.

C. The job fails upon encountering records that violate the expectation.

D. Records violating the expectation are included in the target dataset with an additional flag marking them as invalid.

E. Records violating the expectation are excluded from the target dataset and placed into a quarantine table.

In a Databricks environment, a data engineer needs to ensure their SQL dashboard automatically refreshes daily. However, to optimize resource usage, they aim to limit the SQL endpoint's operation time to only when the refresh is necessary. What steps can the data engineer take to minimize the total active time of the SQL endpoint while still ensuring the dashboard is refreshed once per day?

A. They can ensure the dashboard's SQL endpoint matches each of the queries' SQL endpoints.

B. They can set up the dashboard's SQL endpoint to be serverless.

C. They can turn on the Auto Stop feature for the SQL endpoint.

D. They can ensure the dashboard's SQL endpoint is not one of the included query's SQL endpoint.

What distinguishes the `INSERT OVERWRITE` command from the `CREATE OR REPLACE TABLE` command in Databricks?

A. `INSERT OVERWRITE` maintains historical data versions by default, while `CREATE OR REPLACE` clears the historical data versions by default.

B. `INSERT OVERWRITE` replaces data by default, whereas `CREATE OR REPLACE` replaces both data and schema by default.

C. Both commands are functionally identical and yield the same outcomes.

D. `INSERT OVERWRITE` clears historical data versions by default, while `CREATE OR REPLACE` maintains the historical data versions by default.

E. `INSERT OVERWRITE` replaces data and schema by default, whereas `CREATE OR REPLACE` replaces data by default.

A data engineer is tasked with executing the final segment of a Python script under specific conditions: the variable `day_of_week` must be 1 and `review_period` must be True. Which line of code correctly initiates this segment?

A. if day_of_week = 1 and review_period = “True”

B. if day_of_week == 1 and review_period == “True”

C. if day_of_week = 1 & review_period: = “True”

D. if day_of_week == 1 and review_period

E. if day_of_week = 1 and review_period

A data organization leader is upset about the data analysis team's reports being different from the data engineering team's reports. The leader believes the siloed nature of their organization's data engineering and data analysis architectures is to blame.
Which of the following describes how a data lakehouse could alleviate this issue?

A. Both teams would autoscale their work as data size evolves

B. Both teams would use the same source of truth for their work

C. Both teams would reorganize to report to the same department

D. Both teams would be able to collaborate on projects in real-time

E. Both teams would respond more quickly to ad-hoc requests

In Databricks with Unity Catalog enabled, what determines the physical storage location for a new schema created with:

```
CREATE SCHEMA sample_schema;
```

A. Always dbfs:/user/hive/warehouse

B. The workspace default DBFS root folder

C. Hive metastore's spark.sql.warehouse.dir setting

D. The location defined when creating its parent catalog

E. A random location assigned by Databricks

What happens to the metadata (delta log, history) and the data stored in the storage when you drop an external DELTA table using the SQL command `DROP TABLE table_name`?

A. Drops table from metastore, metadata (delta log, history), and data in storage

B. Drops table from metastore, data but keeps metadata (delta log, history) in storage

C. Drops table from metastore, metadata (delta log, history) but keeps the data in storage

D. Drops table from metastore, but keeps metadata (delta log, history) and data in storage

E. Drops table from metastore and data in storage but keeps metadata (delta log, history)

As a Data Engineer at a retail company utilizing Databricks for data processing, you are tasked with optimizing a Spark SQL query to calculate the total cost for each product in a DataFrame 'df' with columns 'id', 'product', 'quantity', and 'price'. The company mandates applying a 10% discount for 'Electronics' and a 20% discount for 'Groceries'. The solution must not only be cost-effective and scalable for large datasets but also adhere to the company's policy of minimizing computational overhead. Considering these requirements, which of the following queries would you choose to implement? (Choose two correct options.)

A. SELECT product, quantity, price, (quantity * price) AS total_cost FROM df

B. SELECT product, quantity, price, (quantity * price) * CASE WHEN product = 'Electronics' THEN 0.9 WHEN product = 'Groceries' THEN 0.8 ELSE 1 END AS total_cost FROM df

C. SELECT product, quantity, price, (quantity * price) * CASE WHEN product IN ('Electronics', 'Groceries') THEN 0.8 ELSE 1 END AS total_cost FROM df

D. SELECT product, quantity, price, SUM(quantity * price) * CASE WHEN product = 'Electronics' THEN 0.9 WHEN product = 'Groceries' THEN 0.8 ELSE 1 END AS total_cost FROM df GROUP BY product

E. SELECT product, quantity, price, (quantity * price) * CASE WHEN product = 'Electronics' THEN 0.9 WHEN product = 'Groceries' THEN 0.8 ELSE 1 END AS total_cost FROM df UNION ALL SELECT product, quantity, price, (quantity * price) AS total_cost FROM df WHERE product NOT IN ('Electronics', 'Groceries')

In the context of optimizing a data pipeline that utilizes Delta Lake for incremental data processing, consider a scenario where the storage costs are escalating due to the accumulation of deleted files that are no longer needed for the pipeline's operations. The team is looking for a solution that not only addresses the immediate cost concern but also ensures the long-term maintainability and performance of the table. Given these requirements, what is the primary purpose of the `VACUUM` command in Delta Lake, and how does it contribute to the table's maintenance? Choose the best option from the following:

A. The `VACUUM` command is used to reorganize the data in the table to improve query performance by optimizing the physical layout of the data.

B. The `VACUUM` command is used to update the metadata of the table to ensure it accurately reflects the latest data changes and schema modifications.

C. The `VACUUM` command is used to merge small files into larger ones to reduce the number of files in the table, thereby improving the efficiency of file operations.

D. The `VACUUM` command is used to reclaim storage space by permanently removing files that have been marked as deleted and are no longer referenced by the table, thus helping to manage storage costs and maintain table performance.

Which of the following describes the relationship between Gold tables and Silver tables?

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E Gold tables are more likely to contain truthful data than Silver tables.

Your dashboard is loading slowly in the browser because each visualization is processing a large amount of data. Which of the following strategies can effectively address this issue?

A. Expand the maximum scaling range of the SQL endpoint cluster

B. Utilize Delta cache for storing intermediate results

C. Apply Databricks SQL Query filters to reduce data in visualizations

D. Enlarge the SQL endpoint cluster size

E. Delete data from Delta Lake

What is a key advantage of integrating the Unity Catalog with existing systems?

A. It necessitates the use of Databricks-exclusive storage systems.

B. It requires a complete overhaul of existing data catalogs.

C. It operates solely on non-standard SQL variants for legacy systems.

D. It works in concert with existing data, storage, and catalogs, without the need for hard migration.

A data engineer needs to determine whether to use the built-in Databricks Notebooks versioning or version their project using Databricks Repos.
What is an advantage of using Databricks Repos over the Databricks Notebooks versioning?

A. Databricks Repos allows users to revert to previous versions of a notebook

B. Databricks Repos is wholly housed within the Databricks Data Intelligence Platform

C. Databricks Repos provides the ability to comment on specific changes

D. Databricks Repos supports the use of multiple branches

32. Which file in a Databricks Asset Bundle (DAB) allows defining different cluster settings for 'Dev' and 'Prod' environments?

A config.json

B databricks.yml (using targets)

C .gitignore

D requirements.txt

When using the `INSERT OVERWRITE` command to reload the `customer_sales` table, how does it affect the ability to review previous versions of the data?

A. Overwrites the data in the table, all historical versions of the data, you can not time travel to previous versions

B. Overwrites the current version of the data but clears all historical versions of the data, so you can not time travel to previous versions.

C. Appends the data to the current version, you can time travel to previous versions

D. By default, overwrites the data and schema, you cannot perform time travel

E. Overwrites the data in the table but preserves all historical versions of the data, you can time travel to previous versions

5. Which tool in the Databricks UI allows a table owner to easily grant SELECT permissions to other users?

A Dashboards

B Workflows

C Repos

D Catalog Explorer

A data engineer has created a new database using the following command: CREATE DATABASE IF NOT EXISTS customer360;
In which of the following locations will the customer360 database be located?

A. dbfs:/user/hive/database/customer360

B. dbfs:/user/hive/warehouse

C. dbfs:/user/hive/customer360

D. More information is needed to determine the correct response